

Proposed Solution: VayuSangam-AI

Physics-Informed Coupled Weather–Chemistry & Stubble-Plume Intelligence System for Delhi NCR

1. Problem Understanding

Conventional AQI forecasting systems generally use meteorological variables such as temperature, humidity, wind speed and boundary-layer height merely as input features for predicting pollutant concentrations. This ignores an important physical process occurring during severe Delhi-NCR pollution episodes.

Meteorology controls pollutant accumulation and dispersion, but high aerosol concentrations can themselves modify incoming solar radiation, atmospheric temperature, stability and planetary boundary-layer development. During winter, temperature inversions, weak winds and shallow PBL conditions can therefore create a feedback cycle in which pollutants remain trapped near the surface.

The proposed **VayuSangam-AI** system addresses this limitation through an integrated physics + satellite + AI architecture that explicitly models:

Meteorology → Chemistry → Aerosols → Radiation/PBL → Meteorology

while simultaneously tracking external pollution sources such as Punjab-Haryana crop-residue burning.

The system generates high-resolution forecasts of:

- PM_{2.5}
- PM₁₀
- O₃
- NO_x
- Surface AQI
- PBL height
- Atmospheric inversion strength
- Stubble-burning plume movement

for Delhi NCR for the next **0–72 hours**.

2. Core Innovation

Instead of treating AQI forecasting as a pure machine-learning regression problem, VayuSangam-AI consists of three complementary intelligence layers.

Layer 1 — Physics

WRF-Chem performs coupled meteorological and atmospheric-chemistry simulation.

Layer 2 — Earth Observation & Source Intelligence

Satellite observations detect pollutants, aerosols, fires and potential biomass-burning source regions.

Layer 3 — Artificial Intelligence

Machine-learning models correct systematic forecast biases, improve local predictions and provide interpretable forecasts.

Therefore:

WRF-Chem tells us how the atmosphere should behave physically.

Satellites tell us what is actually happening over the region.

AI learns where the physical model consistently deviates from observations.

3. Multi-Source Data Fusion Layer

The system integrates information from multiple satellite, ground and meteorological sources.

A. Sentinel-5P / TROPOMI

Used for atmospheric trace-gas observations including:

- NO₂
- HCHO
- CO
- O₃
- aerosol/cloud-related information

HCHO and NO₂ provide an additional chemical fingerprint for identifying biomass-burning and photochemical activity.

B. NASA FIRMS — VIIRS/MODIS

Used for near-real-time detection of:

- active fires
- thermal anomalies

- fire location
- fire confidence
- Fire Radiative Power (FRP)

Fire detections over Punjab, Haryana, western Uttar Pradesh and neighbouring regions are converted into dynamic stubble-burning source layers.

C. MODIS

MODIS aerosol products provide Aerosol Optical Depth information that can be used for aerosol loading validation and assimilation.

D. INSAT-3D/3DS

Provides regional meteorological and cloud observations useful for monitoring:

- cloud conditions
- surface/atmospheric temperature
- humidity
- radiation-related conditions

E. Meteorological Forecast / Reanalysis

Meteorological boundary and initialization information supplies:

- temperature
- relative humidity
- pressure
- wind speed
- wind direction
- vertical wind
- radiation
- planetary boundary-layer conditions

F. CPCB Ground Monitoring Network

CPCB stations provide surface observations of:

- PM_{2.5}
- PM₁₀
- NO₂
- O₃
- CO
- SO₂

These observations serve two critical purposes:

Ground truth validation

and

ML bias correction.

4. Dynamic Stubble-Burning Emission Intelligence

This module evolves the hotspot ideas used by the earlier BAH teams into an operational source-attribution system.

NASA FIRMS VIIRS/MODIS fire detections are continuously collected for Punjab, Haryana, Rajasthan, Uttar Pradesh and surrounding agricultural regions.

Each detected fire contains:

Location + Time + Fire Radiative Power + Confidence

Fire Radiative Power is used to estimate relative biomass-burning intensity.

The resulting output is a:

Dynamic Stubble Emission Map

Instead of assuming crop-burning emissions are constant, the model updates the external fire contribution according to observed fire activity.

5. HCHO-Assisted Biomass Burning Identification

The HCHO hotspot methodology from the BAH problem is retained as an additional source-identification layer.

Sentinel-5P HCHO observations are spatially processed and combined with:

- VIIRS/MODIS fires
- wind direction
- land-use/agricultural regions
- historical HCHO baseline

Spatial anomaly detection/DBSCAN identifies statistically significant HCHO clusters.

The system then calculates:

Fire-HCHO Spatial Correlation

and

Fire-HCHO Wind Alignment

This helps distinguish probable biomass-burning pollution from unrelated urban chemical hotspots.

The HCHO module therefore changes role:

BAH: HCHO hotspot = final output.

VayuSangam: HCHO hotspot = evidence for pollution-source attribution.

6. WRF-Chem Coupled Atmospheric Engine

WRF-Chem forms the scientific core of the proposed system.

Unlike conventional AQI models where meteorology and pollution are processed independently, WRF-Chem simultaneously simulates atmospheric dynamics and chemistry.

The model produces three-dimensional fields for:

Meteorology

- Temperature
- Wind
- Humidity
- Pressure
- PBL height
- Radiation
- Atmospheric stability

Chemistry

- PM2.5
- PM10
- O₃
- NO_x
- relevant precursor species

Aerosols

Aerosol concentration and optical properties are linked to radiation processes.

Consequently, the system can represent the required feedback chain:

High PM_{2.5}

↓

Higher aerosol loading

↓

Modification of solar radiation

↓

Changes in surface heating

↓

Changes in atmospheric stability / PBL development

↓

Modified pollutant dispersion

↓

Updated PM_{2.5}/O₃

This directly addresses the problem statement's requirement for two-way meteorology–chemistry interaction.

7. Nested High-Resolution Delhi-NCR Modelling

Running an extremely high-resolution model over the whole country would be computationally inefficient.

Therefore nested modelling domains are proposed.

Domain 1 — Regional

North India / Indo-Gangetic Plain

Purpose:

Capture long-range pollution transport and crop-burning emissions.

Domain 2 — Delhi NCR

High-resolution Delhi-NCR domain.

Purpose:

Resolve:

- urban emissions
- local meteorology
- PBL evolution
- inversion events
- PM2.5 gradients
- O₃ formation
- incoming pollution plumes

This allows Punjab/Haryana emissions to enter through the regional domain and subsequently be resolved at much higher spatial resolution over Delhi NCR.

8. Atmospheric Inversion Intelligence Engine

A dedicated inversion module is introduced because inversion is explicitly central to the problem statement.

The module analyses vertical atmospheric profiles rather than relying only on surface temperature.

It considers:

- vertical temperature gradient
- inversion depth
- inversion temperature difference
- PBL height
- surface wind
- vertical mixing
- atmospheric stability

An **Atmospheric Trapping / Inversion Index** is generated.

Example output:

Inversion Strength: SEVERE

PBL Height: 240 m

Wind Speed: 1.1 m/s

Mixing Potential: Very Low

Pollution Trapping Risk: 91%

The system classifies conditions into:

Weak → Moderate → Strong → Severe

9. Stubble Plume Tracking

This module combines the strongest feature of the Observers' HYSPLIT concept with WRF-Chem transport fields.

For every significant agricultural fire cluster, the system determines:

Source

↓

Emission strength

↓

Wind trajectory

↓

Vertical mixing

↓

PBL interaction

↓

Delhi-NCR intersection

↓

Expected PM_{2.5} contribution

HYSPLIT can additionally generate forward/back trajectories for source-attribution visualization.

Therefore, instead of merely displaying:

"Punjab has 1,200 fires."

the system can report:

"A high-intensity Punjab-Haryana plume is forecast to intersect western Delhi between 02:00–06:00 tomorrow."

That is far more actionable.

10. Source Attribution

A major value addition is explaining *why* AQI is deteriorating.

The system estimates the relative influence of:

- crop-residue burning
- local traffic
- industry
- dust
- regional transport
- meteorological trapping

The dashboard can therefore generate explanations such as:

Tomorrow 06:00

Predicted AQI: 421 — Severe

Primary drivers:

- Strong atmospheric inversion
- Low PBL height
- Weak surface winds
- Incoming north-west agricultural plume
- High local PM accumulation

This converts the platform from an AQI number predictor into an atmospheric decision-support system.

11. AI Forecast Correction Layer

The BAH XGBoost/CNN-LSTM work is retained, but AI does **not replace WRF-Chem**.

Instead:

WRF-Chem → AI Bias Correction → Final Forecast

Historical WRF-Chem outputs are aligned with CPCB observations.

Features include:

- WRF PM2.5
- WRF PM10
- WRF O₃
- NO_x
- temperature
- humidity
- wind
- PBL height
- inversion index
- fire intensity
- HCHO
- AOD
- time
- location

Possible models:

XGBoost — primary interpretable correction model.

CNN-LSTM / spatiotemporal model — experimental high-resolution correction model.

The AI learns systematic model biases associated with individual stations, meteorological regimes and pollution episodes.

12. Explainable AI

SHAP-based interpretation is applied to corrected AQI forecasts.

For every prediction the system can show which factors caused AQI to rise.

Example:

Why is tomorrow's AQI predicted as Severe?

- + Strong inversion
- + Incoming crop-burning plume
- + Low PBL
- + Weak wind
- + High initial PM2.5
- Moderate afternoon mixing

This is much more useful to decision-makers than a black-box prediction.

13. 72-Hour Forecast Engine

Forecasts are generated for:

0–24 Hours

Highest-confidence operational forecast.

24–48 Hours

Short-range planning forecast.

48–72 Hours

Early-warning forecast.

Each hour contains:

- AQI
- PM2.5
- PM10
- O₃
- NO_x
- temperature
- wind
- PBL height
- inversion strength
- fire-plume influence

- uncertainty/confidence

This produces an hourly **72-hour atmospheric timeline**.

14. AQI Computation

Predicted pollutant concentrations are converted to AQI using applicable Indian AQI breakpoints.

Instead of predicting AQI directly as an arbitrary ML target:

Atmospheric model → pollutant concentration → corrected concentration → AQI

This preserves physical interpretability.

15. Real-Time Interactive Dashboard

The final system is presented through a GIS-enabled Delhi-NCR dashboard.

Main Map

Interactive high-resolution map showing:

- AQI
- PM2.5
- PM10
- O₃
- NO₂
- PBL height
- HCHO
- active fires
- stubble plume
- wind vectors
- inversion intensity

Users can switch individual layers on/off.

72-Hour Forecast Slider

A timeline allows:

Now → +6 h → +12 h → +24 h → +48 h → +72 h

The user can visually watch pollution entering, accumulating and dispersing across Delhi NCR.

Inversion Panel

Displays:

- inversion strength
- inversion depth
- PBL height
- ventilation conditions
- trapping probability

Stubble Plume Panel

Displays:

- active agricultural fires
- fire intensity
- plume direction
- expected Delhi arrival
- affected NCR districts
- estimated impact level

Station Forecast

Selecting a CPCB station displays hourly forecasts of:

- AQI
- PM2.5
- O₃
- PBL
- inversion
- plume influence

Explainability Panel

Shows the dominant drivers behind each forecast.

16. Early Warning System

Instead of waiting until pollution becomes severe, the system detects dangerous combinations in advance.

For example:

ALERT — Severe Pollution Episode Expected

Time: Tomorrow 02:00–10:00

Predicted AQI: 430–470

Confidence: High

Drivers:

- Strong nocturnal inversion
- PBL below 300 m
- Low wind
- North-west stubble-burning plume
- High existing PM_{2.5}

Such alerts can support:

- pollution-control agencies
 - schools
 - hospitals
 - traffic authorities
 - local administrations
 - vulnerable citizens
-

17. Forecast Uncertainty

The dashboard should not pretend every forecast is exact.

Each prediction is accompanied by:

- confidence score
- expected concentration interval
- model agreement
- data availability indicator

For example:

PM_{2.5} Forecast: 286 µg/m³

Expected range: 255–315 µg/m³

Confidence: 84%

This increases scientific credibility.

18. Model Validation

The system is validated against CPCB monitoring stations.

Metrics include:

- MAE
- RMSE
- R^2
- Mean Bias
- AQI-category accuracy
- severe-event detection rate

Three configurations should be compared:

Baseline

Conventional ML AQI model.

Physics

WRF-Chem forecast.

Proposed Hybrid

WRF-Chem + Satellite + Fire + AI Bias Correction.

The objective is to demonstrate quantitatively that the hybrid system improves both concentration forecasting and severe-pollution-event detection.

19. Counterfactual Analysis — Major USP

A powerful additional feature is the ability to answer:

"What would Delhi's AQI be without today's stubble-burning plume?"

Two model scenarios can be compared:

Scenario A: Full emissions.

Scenario B: Crop-fire emissions removed/reduced.

Difference:

$\Delta\text{PM}_{2.5} = \text{PM}_{2.5} \text{ Full} - \text{PM}_{2.5} \text{ No-Fire}$

This provides an estimate of the stubble-burning contribution during specific episodes.

The same framework can test hypothetical emission-control scenarios.

20. Technology Stack

Atmospheric Modelling

WRF-Chem

WPS

NetCDF

HYSPLIT for trajectory/source-attribution support

Satellite / Earth Observation

Sentinel-5P TROPOMI

NASA FIRMS VIIRS/MODIS

MODIS AOD

INSAT-3D/3DS

Ground Data

CPCB monitoring stations

Data Engineering

Python

Xarray

NetCDF4

GeoPandas

Rasterio

Dask

AI/ML

XGBoost

Scikit-learn

PyTorch/TensorFlow if CNN-LSTM is retained

SHAP

DBSCAN

Backend

FastAPI

PostgreSQL/PostGIS

Redis/cache where required

Dashboard

React/Next.js or Streamlit for rapid prototype

MapLibre/Leaflet

Plotly

21. End-to-End Architecture

Satellite Atmospheric Data

+ NASA FIRMS Fire Data

+ Meteorological Forecast

+ CPCB Ground Stations

+ Emission Inventories

↓

Data Preprocessing & Harmonisation



HCHO + Fire Hotspot Intelligence



Dynamic Biomass-Burning Emissions



Regional WRF-Chem Simulation



Delhi-NCR Nested WRF-Chem Simulation



Meteorology ↔ Chemistry



Aerosol ↔ Radiation/PBL



PM_{2.5} + PM₁₀ + O₃ + NO_x + Meteorology



Inversion Intelligence

+

HYSPLIT / WRF Plume Tracking



XGBoost/CNN-LSTM Bias Correction



AQI Calculation



SHAP Explainability



0–72 Hour Forecast



GIS Dashboard + Early Warning

22. How Previous BAH Ideas Are Combined

SpaceAlpha

Contributes the physics-oriented use of:

TROPOMI + ERA5/PBL + MODIS AOD + FIRMS

This becomes the Earth-observation and atmospheric-context layer.

The Observers

Contributes:

Satellite + CPCB + meteorology + HYSPLIT

HYSPLIT becomes the plume/source-attribution component.

VyomVikram / SAAHAS

Contributes:

CNN-LSTM + satellite + ground + meteorological + fire data

The spatiotemporal ML concept becomes an optional forecast-correction model.

XGBoost Approach

Contributes:

XGBoost + Random Forest comparison

XGBoost becomes the primary physics-model bias-correction layer.

AQI-AI

Contributes:

remote sensing + meteorology + ML + visualization

Its visualization concept becomes part of the operational dashboard.

ASTROTECH

Contributes:

multi-source data fusion + HCHO DBSCAN + high-resolution AQI visualization

DBSCAN/HCHO processing becomes the chemical hotspot intelligence layer.

AntriKSH / Vyom Traya

Contributes:

hotspot + fire + wind alignment

This evolves into fire-source validation and plume-direction intelligence.

None of these modules alone solves the new problem statement.

The key new component that unifies them is:

WRF-Chem coupled weather-chemistry modelling.

23. Key Differentiators

VayuSangam-AI is differentiated from conventional AQI applications through:

1. Two-way weather-chemistry coupling

Not simply weather variables fed into ML.

2. Explicit atmospheric inversion modelling

Forecasts pollution trapping rather than merely detecting high PM.

3. Dynamic stubble-burning plume tracking

Identifies when and where external pollution will reach Delhi.

4. Satellite + Ground + Physics + AI fusion

Uses complementary observations rather than relying on one source.

5. Physics-informed AI

AI corrects the physical model instead of replacing atmospheric science.

6. HCHO-assisted source intelligence

Reuses satellite chemistry to validate probable biomass-burning regions.

7. Explainable forecasts

Explains *why* AQI will deteriorate.

8. 72-hour spatiotemporal forecast

Shows when, where and how pollution will evolve.

9. Counterfactual source analysis

Can estimate how much pollution is associated with external crop-fire emissions.

10. Decision-support dashboard

Transforms atmospheric modelling output into actionable warnings.

24. Proposed Solution in One Line

VayuSangam-AI is a high-resolution, physics-informed 72-hour air-quality forecasting and early-warning system for Delhi NCR that couples WRF-Chem weather-chemistry simulations with satellite observations, real-time crop-fire detection, atmospheric inversion intelligence, plume tracking and explainable AI-based bias correction to predict not only how bad the air will become, but when, where and why the pollution episode will occur.